HYDROFARMS AGRI SOLUTIONS

SIMPLE AUTOMATION, SENSORS, INTERLOCKS AND ALARM SYSTEM

Activity-Specific Installation, Programming and Commissioning Method Statement

Document field	Controlled value
Document code	HF-GH01-MST-AUT-001
Revision	00 - Draft for coordination
Project	One-Feddan NFT Lettuce Greenhouse Project
Location	Alexandria, Egypt
Control philosophy	Distributed simple local control with fail-safe manual fallback
Climate zoning	Four production sectors plus one service zone
Prepared by	Hydrofarms Agri Solutions
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FAIL-SAFE OPERATING RULE
Controller, communications or GSM failure shall alarm but shall not block authorized local Manual-Off-Auto operation. All local protection and emergency stops remain active.

1. Purpose, Scope and Control Philosophy

The project uses distributed local controls, hardwired safety interlocks and GSM alarm reporting rather than a costly full greenhouse SCADA platform. Automatic operation improves consistency while critical fan and pump groups remain locally operable.

Subsystem	Control assembly	Controlled scope
Production climate control	1 integrated control section	35 exhaust fans in four sectors 9+9+8+9; 28 circulation fans; production T/RH sensors.
Service climate control	1 integrated control section	Three exhaust fans staged 1+2; two circulation fans; one service T/RH sensor.
Cooling-pad pump control	1 integrated control section	Four production water circuits and one service circuit; level, pump and fan permissives.
NFT/manoeuvre pump control	1 integrated control section	Production, nursery and reserve-transfer duty/standby logic, dry-run and chiller permissives.
Germination control	1 local panel	Independent pump control, misting timer and grow-lighting timer.
Alarm communication	1 GSM alarm transmitter	Critical temperature, level, pump, phase and common supplier-package alarms.

Anti-duplication: power DBs, starters, feeders and control cables belong to electrical works. Supplier packages retain their own internal safety and control; only external interfaces are coordinated here.

2. Applicable Standards and Submittals

- IEC 61131-2 for industrial control equipment and compact PLC/smart-relay hardware.
- IEC 60204-1 for machinery electrical/programmed control equipment.
- IEC 61326-1 for control/instrument EMC and IEC 60529 for IP ratings.
- IEC 60364/61439, Egyptian requirements and approved supplier instructions.

Submittals: panel schematics, I/O schedule, sensor layout, alarm list, control narratives, interface matrix, terminal schedule, program backup and FAT/SAT procedures.

3. Minimum Field Devices and Quantities

Code	Device	Qty	Minimum requirement
CTRL-006	Production T/RH sensors	4	Industrial combined sensors, one per production climate sector; IP65 remote head or protected transmitter.
CTRL-007	Service T/RH sensor	1	Combined sensor located in representative occupied/crop air, away from direct pad spray.
CTRL-008	Outdoor temperature sensor	1	Shielded north-facing ambient reference, clear of condenser/fan discharge and solar radiation.
CTRL-009	Nutrient temperature interfaces	3	One value/status per east, west and nursery tank from chiller package; no duplicate PT100 hardware.
CTRL-010	Low-level float switches	8	Four reserve, two production, one nursery and one germination tank; counterweighted or guided float.
CTRL-011	High-level float switches	8	Same eight tanks; independent stop/alarm contact at the approved high level.
CTRL-012	Backup thermostats	2	Independent adjustable thermostats for production and service high-temperature fail-safe control.
CTRL-013	Germination misting timer	1	Cycle timer with manual override; controls local pump/solenoid through interposing relay.
CTRL-014	Germination lighting timer	1	24-hour/weekly timer with manual override and retained programme.
CTRL-015	Manual-Off-Auto selectors	28	Panel-mounted selector switches for principal groups and duty/standby functions.
CTRL-016	Run/fault indicator lamps	56	Twenty-eight green run and twenty-eight red fault indicators.
CTRL-017	Interposing control relays	30	Plug-in relays with LED and test button, spare contacts and labeled bases.
CTRL-018	Field junction boxes	6	IP65 minimum, gland plate, terminals, earth stud and 20% spare terminals.
CTRL-019	GSM alarm unit	1	Minimum four recipients, event text, power-fail alarm and battery backup.
CTRL-020	Aspirated/radiation shields	5	Four production sensor shields and one outdoor shield; service sensor protected from splash/drafts.

4. Supplier-Package Interface Boundary

Package/system	Ownership	Coordinated signals
Five 7.5 TR chillers	Supplier package	Enable/permissive, common run, common fault, tank temperature and high/low temperature alarm.
Geneina fertigation	Supplier package	Common run/fault, low stock, EC/pH alarm and emergency stop status; no external dosing logic.
Water-treatment station	Supplier package	Remote enable, common run/fault, product-water available and reserve-bank high-level inhibit.
Twenty screen drives	Screen supplier package	Common enable/close command, common fault, position/status and wind/rain alarm; local limits remain supplier scope.
Pad-water tanks	Cooling package	Five low/high level and pump-status dry contacts; physical switches are not duplicated here.
Electrical power system	Electrical chapter	Power feeders, starters/MPCB, isolators, control cable and earthing; control contacts land at labeled terminals.

No external command may bypass a supplier safety trip, local emergency stop or motor protection.

5. Production Climate Sequence

Stage	Initial trigger	Action
S0	Below 26 degC	Circulation fans only; log T/RH; pads dry.
S1	26 degC	Start first exhaust-fan sector/group for dry ventilation.
S2	28 degC	Start required pad-pump sector, prove level/flow, pre-wet 30-60 s, then add next fan group.
S3	30 degC	Add third fan group/sector while monitoring RH and pump status.
S4	32 degC	All available production fan sectors and required pad circuits operate.
ALM-H	35 degC	High-temperature audible/GSM alarm; force maximum available cooling unless a safety trip prevents it.
ALM-HH	38 degC	Critical alarm and operator escalation; maintain safe ventilation and inspect airflow/water/power.

Initial triggers are adjustable at commissioning. Use sector values for local staging and the highest valid temperature for override/alarm, with approximately 1 degC hysteresis and minimum run timers.

5.1 Relative-humidity restraint

Condition	Control response
RH below 85%	Normal staged pad operation permitted.
RH 85-90%	Maintain ventilation; delay or pulse additional pad sectors where temperature allows.
RH above 90%	High-RH warning; inhibit unnecessary pad stages but retain dry ventilation.
High temperature override	At 35 degC or higher, the configured safety override may permit required pad cooling despite high RH; event is logged.

6. Service-Greenhouse Climate Sequence

Stage	Initial trigger	Action
S0	Below 26 degC	Two circulation fans as scheduled; exhaust off unless minimum ventilation is required.
S1	26 degC	Start one exhaust fan for dry ventilation.
S2	28 degC	Start service pad pump, prove water, pre-wet 30-60 s, then run two exhaust fans.
S3	30 degC	Run all three service exhaust fans with pad cooling.
ALM	35 degC	High-temperature audible/GSM alarm and maximum available cooling.

7. Pump, Tank and Chiller Logic

Service	Arrangement	Control/interlock
Production NFT	Two duty pumps plus one common standby	Enable only above low level; duty pump run feedback required; standby selection cannot cross-connect east and west tanks.
Nursery NFT	One duty plus one standby	Automatic changeover on duty trip; low level blocks both; alarm if standby fails.
Reserve transfer	One duty plus one standby	Fill one operating tank at a time; high-level stops transfer; valve line-up and tank selection must agree.
Production pad pumps	Four duty sectors plus warehouse spare	Low pad-tank level stops the related pump; fans remain available for dry ventilation.
Service pad pump	One duty plus installed/warehouse standby basis	Low level blocks cooling stage; service fans remain available for dry ventilation.
Germination pump	One local pump	Timer/manual operation; independent low-level trip at the 1 m3 germination tank.

Function	Set point/permissive	Action
Control target	20 degC	Normal adjustable band 19-21 degC.
Enable	Safe tank level + NFT pump run/flow permissive	No refrigeration without solution circulation.
High alarm	23 degC	Common chiller alarm and GSM escalation if persistent.
Critical high	25 degC	Critical crop/process alarm; verify all refrigeration circuits and circulation.
Local low cut-out	17 degC	Independent supplier-package shutdown; central control cannot override.
Lead/lag	Two 7.5 TR units per production tank	Stage second unit after adjustable delay; alternate lead weekly or by runtime.

8. Screen, Fertigation and WTP Coordination

- Screen supplier controls 15 external and five internal drives, limits, 20 isolator/E-stops, two wind sensors and one rain sensor; do not duplicate.
- Central control may issue common enable/safe-close. Wind/rain/overload actions remain local to the screen package.
- Geneina alone controls A/B/acid dosing and EC/pH; central automation receives alarm/status only.
- WTP package controls treatment/flushing; central control provides enable and reserve-bank high-level inhibit.

9. Alarm and GSM Matrix

Condition	Automatic response	Source	Reset/priority
Production/service high temperature	Audible + beacon + GSM	Cooling controller	Immediate
Nutrient solution high temperature	Panel + GSM	Chiller interface	Immediate
Operating/reserve/germination low level	Stop affected pump + panel + GSM	Float switch	Latched
Tank high level	Stop related fill/transfer + panel	Float switch	Latched
Pad-tank low level	Stop related pad pump; dry ventilation remains	Cooling package	Latched
Pump/fan motor trip or phase fault	Stop affected group + alarm; start standby where available	Electrical starter	Latched
Sensor failed/out of range	Alarm and use valid sensors/backup thermostat	Controller diagnostics	Latched
GSM/power failure	Local alarm retained; battery power-fail message	GSM unit	Immediate
Supplier package common fault	Panel + GSM according to priority	Dry contact/Modbus	Latched

Alarm texts identify greenhouse, system and zone/tank. GSM sends to at least four approved recipients, reports mains failure using battery backup and retains dated event history.

10. Installation and Programming Sequence

- Approve the control philosophy, panel schematics, I/O schedule, sensor layout, interface matrix, alarm list and cause-and-effect before fabrication.
- Bench-test all five control sections, GSM unit, selectors, lamps, relays, timers and simulated field inputs before dispatch.
- Install panels in dry accessible positions, maintaining electrical clearances and separation from chemical transfer, pad spray, direct sun and fan discharge.
- Install four production T/RH sensors at representative crop height along the air path and climate sectors; install the service sensor in the germination/packing air zone without direct pad spray.
- Install the outdoor sensor in its radiation shield on a north-facing ventilated location clear of condensers, walls and exhaust discharge.
- Install tank floats at surveyed set levels with accessible withdrawal, no contact with suction turbulence, and individual cable/terminal identification.
- Route instrument and control cables separately from power cables. Use shielded cable for analogue/communications signals, earth the shield at one designated end, and seal all field entries.
- Terminate every supplier interface through labeled dry-contact/analogue/Modbus terminals. Do not modify vendor internal safety circuits or bypass local emergency stops.
- Complete point-to-point checks, sensor calibration, timer settings, alarm-recipient setup and cause-and-effect simulation before wet commissioning.

10.1 Programming and access control

- Operator and Engineer access levels; protected logic/set points for engineers only.
- Dated native program backup, parameter list and version record before and after approved changes.
- Manual mode never bypasses overload, phase loss, low-level, E-stop or supplier safety.
- Staggered automatic restart after power restoration; no simultaneous uncontrolled motor start.

11. Testing, Commissioning and Acceptance

Test	Method	Acceptance
Panel FAT	Simulate every input/output, selector, lamp, relay, timer and alarm	Approved FAT sheet; no unexplained action
Point-to-point	Trace each field cable and interface terminal	100% correct identification and polarity
T/RH calibration	Compare with traceable reference at installed points	Temperature within +/-0.5 degC; RH within +/-3% RH
Tank temperature interface	Compare central display with local chiller/reference value	Within +/-0.5 degC
Float switches	Raise/lower each device through trip points	Correct stop, alarm and reset; no cross-action
Climate stages	Heat/simulate set points S0-S4 and service S0-S3	Correct sequence, delay, fan/pad permissives
Duty/standby	Trip each duty pump and simulate low level	Standby starts only where permitted; dry-run prevented
Chiller interlock	Remove NFT permissive and simulate temperature alarms	Refrigeration inhibited; local safety retained
Screen interface	Wind/rain/close/fault simulation	Supplier package responds and reports without duplicate control
GSM	Test all critical alarms and mains failure	Correct text delivered to all approved recipients
Manual fallback	Disable automatic controller/communication	Authorized local manual operation remains available and safe
Integrated run	Eight-hour summer-mode operation	Stable sequencing, no nuisance trip, complete event log

11.1 Inspection and Test Plan

No.	Inspection stage	Method	Type	Record
01	Control philosophy, I/O and cause/effect	Document review	H	Approved design pack
02	Panels and programmed logic	FAT	W	FAT report
03	Sensor/float locations before cable closeout	Survey/visual	H	Installation release
04	Cabling, shielding, terminals and labels	Continuity/visual	W	Point schedule
05	Sensor and timer calibration	Instrumented	H	Calibration sheets
06	Motor, level and supplier-package interlocks	Functional	H	Cause/effect record
07	Climate and pump sequences	Performance	H	SAT report
08	GSM alarms and power failure	Functional	W	Alarm report
09	Integrated eight-hour run	Performance	H	Run log
10	Backups, as-builts, training and handover	Record review	R	Turnover dossier

Intervention codes: H = Hold, W = Witness, S = Surveillance, R = Record review.

11.2 Hold points

- HP-AUT-01: Narrative, I/O, interfaces and cause/effect accepted before fabrication.
- HP-AUT-02: FAT accepted before dispatch.
- HP-AUT-03: Installation and point-to-point accepted before wet commissioning.
- HP-AUT-04: Calibration and safety interlocks accepted before Auto mode.
- HP-AUT-05: Climate, pump, supplier-interface, GSM and manual-fallback SAT accepted before handover.

12. HSE, Records and Handover

Hazard	Mandatory control	Residual risk
Unexpected start	LOTO, inhibit outputs, local isolation and test announcement	Low
Live panel	Authorized personnel, insulated tools and test-before-touch	Medium
Wet greenhouse	IP65 equipment, sealed glands, drip loops and bonding	Low
Work at height	Approved access and exclusion zone	Medium
Manual override	Training, labels, protected settings and logged operation	Low
Communication loss	Local alarm and manual operation remain available	Low

Handover includes schematics, I/O and terminal schedules, program backups/passwords, parameter register, calibration, FAT/SAT, cause/effect, GSM tests, alarm logs, as-builts, manuals, warranties, spares and training.